Anticipatory Action for Kenya's Compound Climate-Food-Conflict Crisis

A 2-3 month early warning window exists. It is not being used.

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47	19	7
counties analysed	**years of data**	**high-risk counties**
All Kenyan counties	2006 to 2025	MPI ≥ 0.30 (ASAL)

The Problem

Kenya loses an estimated 2-3% of GDP annually to drought-related crises. The 2011 Horn of Africa famine, the 2017 ASAL drought, and the 2022 La Nina event each triggered national emergencies. Each one followed the same sequence: rainfall failed, food prices spiked, and intercommunal conflict escalated.

What is missing is not data. Kenya has 19 years of satellite-derived rainfall records, humanitarian price monitoring, and conflict event data at county level. What is missing is a system that reads these signals together, in sequence, and in time to act.

This brief presents findings from an independent analysis of 19 years of humanitarian data across 47 Kenyan counties. It demonstrates that the rainfall-price-conflict sequence is statistically detectable and that a measurable early warning window exists before each stage of the cascade.

What the Analysis Found

A directed cascade, conditional on structural vulnerability

Rainfall deficits in Kenya's arid and semi-arid land (ASAL) counties systematically precede real food price spikes, which in turn precede intercommunal conflict escalation. This sequence does not appear in well-integrated urban counties. It is specific to counties where structural poverty, market isolation, and climate exposure combine.

The cascade was tested using Augmented Dickey-Fuller stationarity tests, cross-correlation functions, and Granger causality tests across three counties with sufficient contiguous food price data: Marsabit, Turkana, and Mandera.

County	Link tested	p-value	Lag	Result
Marsabit	**Rainfall → Price**	**0.040**	**2 months**	**Significant**
Marsabit	**Price → Conflict**	**0.021**	**1 month**	**Significant**
Turkana	Rainfall → Price	0.077	3 months	Marginal
Turkana	Price → Conflict	0.073	1 month	Marginal
Mandera	Rainfall → Price	0.499	1 month	Not significant (cross-border dynamics)
Mandera	Price → Conflict	0.854	4 months	Not significant

Reverse direction tested for all links. Non-significant in all counties. These results are exploratory: approximately 90 tests were run across all counties, lags, and directions. No result survives Bonferroni correction (threshold p < 0.0006). The cascade interpretation rests on directional consistency across counties and cross-correlation patterns, not on any individual p-value.

The cascade lives in seven counties

The two-link cascade is absent in medium and low-vulnerability counties. It is detectable only in the seven ASAL counties with a Multidimensional Poverty Index (MPI) at or above 0.30 (Oxford OPHI 2022 survey). These counties share three characteristics: high climate exposure, limited market integration, and household dependence on rain-fed pastoral livelihoods.

Rank	County	MPI	Tier
1	Turkana	0.50	High
2	Mandera	0.46	High
3	Samburu	0.42	High
4	Wajir	0.40	High
5	Tana River	0.38	High
6	West Pokot	0.35	High
7	Marsabit	0.35	High

MPI source: Oxford Poverty and Human Development Initiative (OPHI), 2022 survey. County poverty rankings assumed stable over the 2009-2020 analysis window.

Each drought raises the price floor permanently

Real maize prices in ASAL counties did not return to pre-shock levels after the 2011 famine or the 2017 drought. Each drought cycle permanently raised the price floor. Households entering the 2022 La Nina event were already operating at a higher real price baseline than at any prior point in the series. Single-event analysis systematically understates cumulative stress.

The Early Warning Window

The cascade structure creates a predictable operational window for anticipatory action:

Month 0
Rainfall deficit confirmed
Below-normal long-rains season detected in ASAL counties via CHIRPS monitoring.

Month 2-3
Price monitoring trigger
Activate market price monitoring in 7 high-risk counties. Alert county food security committees.

Month 5-7
Conflict early warning
Activate conflict early warning and pre-position protection resources in ASAL counties.

Month 8+
Emergency response
Cascade has likely materialised. Emergency response is reactive, not anticipatory.

Key finding: A below-normal long-rains season in March should trigger price monitoring by May-June and conflict early warning by July-August in the 7 high-risk counties. The data infrastructure to detect it already exists. What is missing is integration of these signals into a single sequential monitoring framework.

Recommendations for NDMA

1. Integrate rainfall and price monitoring into a sequential trigger

NDMA already monitors both rainfall anomalies and food prices. The missing step is treating rainfall deficits as a leading indicator for price monitoring activation. A below-normal long-rains season (RFQ < 80%) in any of the 7 high-risk counties should automatically trigger enhanced price monitoring for that county in the following 2-3 months.

2. Target anticipatory resources at the 7 counties

National drought declarations distribute resources across all 47 counties. The cascade mechanism operates only in the 7 ASAL counties with MPI ≥ 0.30. Geographically undifferentiated responses will always underestimate need in these counties while over-allocating to counties where the cascade does not operate.

3. Adopt a cumulative stress lens for repeat drought events

Emergency response architecture is designed for single events. The ratchet effect identified in this analysis means that households entering each successive drought cycle are more vulnerable than in prior cycles, because real price floors have risen permanently. NDMA's vulnerability assessments should incorporate multi-cycle cumulative stress, not only current-event severity.

4. Close the 2021-2025 food price data gap

WFP VAM price monitoring coverage ends 2020-2021 for most ASAL counties. The 2022 La Nina drought shows clear rainfall deficits and conflict escalation but price transmission cannot be confirmed due to missing data. Restoring systematic retail price monitoring in the 7 high-risk counties would enable real-time cascade detection.

Limitations

These findings are exploratory. Granger tests were run across approximately 90 county-lag-direction combinations. No result survives Bonferroni correction (adjusted threshold $p < 0.0006$). The cascade interpretation therefore rests on directional consistency across counties and cross-correlation patterns, not on any individual p-value. Replication with larger samples, additional counties, or panel Granger methods (such as Dumitrescu-Hurlin) would be required to confirm these findings.

This analysis confirms temporal precedence, not structural causation. Granger tests establish that lagged rainfall deficits improve prediction of price changes, and lagged price changes improve prediction of conflict events. Unmeasured confounders including governance quality, humanitarian aid flows, and cross-border trade dynamics may independently drive both outcomes. The household-level behavioural mechanism connecting price stress to conflict participation is not resolved by this aggregate analysis.

Granger analysis was restricted to three counties with sufficient contiguous food price data (Marsabit, Turkana, Mandera). Results do not generalise statistically to all seven high-risk counties, though the Vulnerability Risk Index and cross-correlation analysis cover all 47 counties.

Data and Code

All analysis is publicly available and reproducible. Five Jupyter notebooks cover data coverage auditing, cleaning, exploratory analysis, lag correlation testing, and the Vulnerability Risk Index construction. All data is sourced from the HDX Humanitarian API.

- GitHub: github.com/MutwiriC/compound-crisis-cascade-kenya
- Data: data.humdata.org/dataset/hdx-hapi-ken
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This brief is submitted for consideration by the National Drought Management Authority. The author welcomes feedback, collaboration, and the opportunity to present these findings to NDMA's technical working group.